Department of Mathematics and Computer Science
NATIONAL UNIVERSITY OF LESOTHO

M3503 **Tutorial 6** **October 2025**

1. Use the elimination method to find a general solution for the given linear system, where differentiation is with respect to t

(a) $x' + 2y = 0,$
$x' = x - y,$

(b) $x' - y' = 0,$
$y' = y - 4x$

(c) $x' + y' - x = 5,$
$x' = 3x - 2y + \sin t$

(d) $x' + y' + y = 1$
$y' = 4x - y - \cos t$

(e) $(D+1)[u] - (D+1)[v] = e^t,$
$(D-1)[u] + (2D+1)[v] = 5$

(f) $(D-3)[x] + (D-1)[y] = t$
$(D+1)[x] + (D+4)[y] = 1$'

(g) $x' + y' + 2x = 0,$
$x' + y' - x - y = \sin t$

(h) $2x' + y' - x - y = e^{-t},$
$x' + y' + 2x + y = e^t$

(i) $(D^2 - 1)[u] + 5v = e^t,$
$D^2[u] + D[v] = 2$

(j) $2u + (D^2 + 2)[v] = 0$
$4u + D[v] = 6$

(k) $\dfrac{dx}{dt} = x - 4y,$
$\dfrac{dx}{dt} + y = t^2$

(l) $\dfrac{dy}{dt} = x + y,$
$-x + \dfrac{dy}{dt} = 1$

(m) $\dfrac{dw}{dt} = 5w + 2z + 5t,$
$\dfrac{dz}{dt} = 3w + 4z + 17t$

(n) $\dfrac{dx}{dt} + x + \dfrac{dy}{dt} = e^{4t},$
$2x + \dfrac{d^2y}{dt^2} = 0$

(o) $x'' + 5x - 4y = 0,$
$-x + y'' + 2y = 0$

(p) $x'' + y'' - x' = 2t,$
$x'' + y' - x + y = -1$

2. Find the general solution of the given system, by first converting the given system into eigenvalue-problem

(a) $\dfrac{dx}{dt} = x + 2y$
$\dfrac{dy}{dt} = 4x + 3y$

(b) $\dfrac{dx}{dt} = -4x + 2y$
$\dfrac{dy}{dt} = -\dfrac{5}{2}x + 2y$

(c) $\mathbf{X}' = \begin{pmatrix} 10 & -5 \\ 8 & -12 \end{pmatrix} \mathbf{X}$

(d) $\dfrac{dx}{dt} = x + y - z$
$\dfrac{dy}{dt} = 2y$
$\dfrac{dz}{dt} = y - z$

(e) $\dfrac{dx}{dt} = 3x - y$
$\dfrac{dy}{dt} = 9x - 3y$

(f) $\mathbf{X}' = \begin{pmatrix} -1 & 3 \\ -3 & 5 \end{pmatrix} \mathbf{X}$

(g) $\dfrac{dx}{dt} = 3x - y - z$
$\dfrac{dy}{dt} = x + y - z$
$\dfrac{dz}{dt} = x - y + z$

(h) $\dfrac{dx}{dt} = 6x - y$
$\dfrac{dy}{dt} = 5x + 2y$

(i) $\dfrac{dx}{dt} = 5x + y$
$\dfrac{dy}{dt} = -2x + 3y$

(j) $\mathbf{X}' = \begin{pmatrix} 4 & -5 \\ 5 & -4 \end{pmatrix} \mathbf{X}$

3. Solve the given initial-value problem.

(a) $\mathbf{X}' = \begin{pmatrix} 1 & -12 & -14 \\ 1 & 2 & -3 \\ 1 & 1 & -2 \end{pmatrix} \mathbf{X}, \quad \mathbf{X}(0) = \begin{pmatrix} 4 \\ 6 \\ -7 \end{pmatrix}$

(b) $\mathbf{X}' = \begin{pmatrix} 6 & -1 \\ 5 & 4 \end{pmatrix} \mathbf{X}, \quad \mathbf{X}(0) = \begin{pmatrix} -2 \\ 8 \end{pmatrix}$

4. Show that the 5×5 matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

has an eigenvalue λ_1 of multiplicity 5 . Show that three linearly independent eigenvectors corresponding to λ_1 can be found.